

Benchmarking Transformers and Baselines for Multi-Horizon Stock Return Prediction

Technical and Earnings Features on the Dow 30

Nelson Siu¹ Jonathan H. Chan²

¹Division of Engineering Science, University of Toronto

²Innovative Cognitive Computing, King Mongkut's University of Technology Thonburi

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Why is this hard?

- Noisy, low signal-to-noise ratio
- Regime-dependent patterns
- Market efficiency limits
- Non-stationarity

Why does it matter?

- Trading strategies
- Risk management
- Portfolio optimization
- Understanding market dynamics

Key Question

Do complex transformer models outperform simpler baselines on daily stock data?

Our Approach

Rigorous benchmarking across:

- Multiple horizons (1, 5, 21 days)
- Multiple model classes
- Multiple market regimes

What's missing in current literature?

- ❶ **Fair comparisons:** Most studies don't compare transformers, RNNs, and tabular models on identical features
- ❷ **Event features underutilized:** Earnings and event data rarely integrated into deep learning models
- ❸ **Look-ahead bias:** Many protocols risk data leakage
- ❹ **Regime analysis:** Limited evaluation across different market conditions

Our Contribution

Apples-to-apples comparison with strict temporal splits, fixed universe (Dow 30), and comprehensive regime + event analysis

① Robust experimental design

- Strict temporal split: train 2016–2019, val 2020, test 2021–2024
- Fixed Dow 30 constituents (no survivorship bias)
- Forward-only validation with 5-day embargos

② Fair model comparison

- Identical features across all models
- Multiple model classes: RNN, Transformer, Tabular
- Three random seeds for statistical reliability

③ Comprehensive evaluation

- Multi-metric: RMSE, R^2 , Directional Accuracy
- Regime analysis: bear (2022) vs. rally (2023–24)
- Event windows: earnings announcements

④ Surprising insights

- Simple models can match or beat complex ones
- Event features matter significantly
- Performance is highly regime-dependent

Three Core Hypotheses

H1: Model Class Advantage

Hypothesis: Temporal Fusion Transformer (TFT) will outperform baselines at $h = 21$ days

Rationale: Attention mechanisms should handle long-term dependencies better

Criteria: ≥ 2 pp improvement in DA or $\geq 5\%$ reduction in RMSE

H2: Event Features Add Value

Hypothesis: Adding earnings features improves DA without worsening RMSE

Rationale: Post-earnings-announcement drift is well-documented

Criteria: DA $\uparrow$ by ≥ 2 pp while RMSE stable

H3: Regime Dependence

Hypothesis: Model rankings differ between bear and bull markets

Rationale: Adaptive Markets Hypothesis suggests pattern instability

Criteria: Different best-performing models across 2022 bear vs. 2023–24 rally

Traditional Approaches

- **ARIMA/GARCH:** Linear, assumes stationarity
- **Ridge Regression:** Simple, interpretable
- **Random Forest/XGBoost:** Strong with engineered features

Pros: Fast, interpretable, less prone to overfitting

Cons: Miss temporal patterns, degrade at longer horizons

Deep Learning Approaches

- **LSTM/GRU:** Capture temporal dependencies
- **CNN-LSTM:** Extract local + global patterns
- **Transformers (TFT):** Attention-based, multi-horizon

Pros: Handle sequences, learn complex patterns

Cons: More parameters, require more data, prone to overfitting

Key Insight from Literature

Model performance is highly task- and data-dependent. **No universal winner.**

Transformers revolutionized NLP and time-series forecasting

- **PatchTST:** Patch-wise approach improves long-horizon forecasting
- **TimesFM:** Large-scale pretraining enables zero-shot forecasting
- **Financial LLMs:** BloombergGPT, FinGPT integrate text + prices

The TFT Architecture

- Static covariate encoders
- Variable selection networks
- Multi-head attention
- Multi-horizon prediction heads
- Interpretable via attention weights

Question: Do these advantages hold on *small, daily financial panels*?

`figs/experiment_timeline.png`

Feature Groups

1. Technical Indicators

- Momentum (5, 10, 20 day)
- Moving averages
- Volatility (rolling std)
- RSI, MACD
- Volume features

2. Earnings Features

- Earnings surprise %
- Days to/from earnings
- Announcement flags
- Event windows (± 3 days)

Data Sources

- **Prices:** yfinance
- **Earnings:** API Ninjas
- **Macro:** FRED API

Leakage Controls

- After-market data shifted to $t + 1$
- Forward-only blocks
- 5-day embargos between splits
- Scaling fit on train only
- No look-ahead bias

Global Training

- Pooled across all 30 stocks

Tabular Models

- **Ridge Regression**
 - Linear baseline
 - L2 regularization
 - α tuned on validation
- **Random Forest**
 - 100 trees, depth 5
 - Strong tabular learner
- **XGBoost**
 - Gradient boosting
 - Depth 3, lr 0.1

Trained separately per horizon

Sequence Models

- **LSTM**
 - 64 hidden, 1 layer
 - 60-day lookback
 - Multi-horizon output
- **GRU**
 - 64 hidden, 1 layer
 - Simpler than LSTM
- **TFT (Temporal Fusion Transformer)**
 - 4 attn heads, 1 layer
 - Hidden dim 16
 - Multi-horizon output
 - ~50k parameters

Predict all horizons simultaneously

`figs/tft_architecture.png`

Three complementary metrics:

① **Root Mean Squared Error (RMSE)**

- Measures prediction precision in log-return units

② **Coefficient of Determination (R^2)**

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i y_i^2}$$

- vs. zero-return baseline
- Negative values common (low signal-to-noise)

③ **Directional Accuracy (DA)**

$$DA = \Pr[\text{sign}(\hat{r}) = \text{sign}(r)]$$

- Most important for trading
- 50% = random, >50% = predictive

Main Results: Performance by Horizon

Model	h	RMSE	R^2	DA
<i>$h = 1$ (next day)</i>				
LSTM	1	0.0162	-0.003	51.7%
GRU	1	0.0162	-0.006	50.1%
Rand. Forest	1	0.0164	-0.023	51.9%
Ridge	1	0.0163	-0.018	50.0%
<i>$h = 5$ (one week)</i>				
Ridge	5	0.0365	-0.020	51.2%
Rand. Forest	5	0.0397	-0.207	53.1%
LSTM	5	0.0578	-0.002	55.4%
GRU	5	0.0579	-0.003	54.2%
<i>$h = 21$ (one month)</i>				
LSTM	21	0.0508	-0.002	55.0%
GRU	21	0.0509	-0.002	53.9%
TFT	21	0.0510	-0.007	53.9%
Ridge	21	0.0742	-0.028	52.8%

Key Takeaway

LSTM wins at $h = 1$ & $h = 21$. **Ridge** dominates at $h = 5$. TFT competitive but not superior.

`../comparison/artifacts/figures/fig1_da_by_horizon.png`

Why Are All R^2 Values Negative?

This is Normal in Finance!

$$R^2 \text{ vs. zero: } R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum y_i^2}$$

Interpretation:

- Negative R^2 = worse than predicting zero
- Reflects market efficiency (low signal/noise)
- **LSTM (-0.002)** better
- **Rand. Forest (-0.248)** overfitting

Important

Despite negative R^2 , **RMSE** and **DA** show meaningful differences. *DA is 50% economically significant.*

Hypothesis 1: TFT Superiority?

H1: Model Class Advantage

Hypothesis: TFT will outperform baselines at $h = 21$

Criteria: ≥ 2 pp DA improvement or $\geq 5\%$ RMSE reduction

Results at $h = 21$:

Model	RMSE	DA (%)
LSTM	0.05084	55.0
GRU	0.05085	53.9
TFT	0.05096	53.9

Verdict: H1 NOT SUPPORTED

- LSTM beats TFT by 0.24% RMSE and 1.1 pp DA
- Differences $<$ thresholds (2 pp, 5%)
- **Attention provides no advantage here**

Hypothesis 2: Earnings Features Matter?

Ablation Study at $h = 21$

Features	RMSE	DA
<i>TFT</i>		
Tech only	0.0513	52.2%
Tech+Earnings	0.0509	53.3%
<i>XGBoost</i>		
Tech only	0.0817	51.1%
Tech+Earnings	0.0812	51.8%

Overall improvement:

- DA: +1.1 pp
- RMSE: slightly better

Earnings Window Analysis

Period	RMSE	DA
Earnings ($\pm 3d$)	0.0695	61.4%
Non-earnings	0.0666	53.9%

During earnings windows:

- +7.5 pp improvement in DA!
- Modest RMSE increase (higher volatility)

Hypothesis 3: Regime Dependence

../comparison/artifacts/figures/fig2_regime_rmse_heatmap.png

Best Models by Regime

Regime	Best	RMSE
<i>Bear 2022</i>		
	GRU	0.0636
<i>Rally 2023–24</i>		
	LSTM	0.0475

Directional Accuracy

● Bear 2022: All <50%

● Rally: 55.2%

Why Did Simple Models Win?

① Data constraints

- Small panel: 30 stocks, daily frequency
- Limited sample for deep learning; TFT's 50k params may be too many

② Signal characteristics

- Very low signal-to-noise ratio
- Short-term patterns quasi-linear (Ridge wins at $h = 5$)

③ Model capacity trade-offs

- More parameters $\Rightarrow$ harder training with weak signals
- LSTM/GRU ($\sim 25k$) vs. TFT ($\sim 50k$ params)
- Simpler = better bias-variance trade-off

④ Feature engineering matters

- Strong technical indicators capture most signal
- Tabular models exploit these effectively

When Do Transformers Shine?

Our negative result for TFT suggests:

Transformers Need More

① Larger datasets

- TimesFM: trained on 100B time points
- Our study: ~ 30 stocks $\times$ 8 years

② Richer modalities

- Text: news, earnings calls, social media
- Alternative data: satellite, credit card
- BloombergGPT, FinGPT integrate multimodal data

③ Longer horizons / higher frequency

- Intraday: more samples; Long-term: clearer patterns

④ Pretraining

- Transfer learning; Foundation models for finance

Model Selection by Horizon

- $h = 1$ (**daily**): LSTM or Random Forest
- $h = 5$ (**weekly**): Ridge regression
- $h = 21$ (**monthly**): LSTM or GRU

When to Use Transformers

Only if you have:

- Large datasets (many stocks, long history)
- Multimodal inputs (text + prices)
- Computational resources
- Ability to leverage pretraining

Feature Engineering Matters

Important Caveat

We focus on **statistical accuracy** (RMSE, R^2 , DA), not **profitability**

What we don't test:

- Trading strategies, transaction costs
- Position sizing, risk management
- Sharpe ratio, maximum drawdown

Reality Check

- 55% directional accuracy *might* be profitable
- Depends on: trading frequency, spreads, commissions
- Full backtesting needed for economic value

- ❶ **Limited universe:** Dow 30 only (large-cap U.S.)
 - May not generalize to small-cap, international, or other assets
- ❷ **Daily frequency only**
 - Intraday patterns not explored
 - Lower frequencies (weekly, monthly) not tested
- ❸ **Hyperparameter tuning**
 - Lightweight tuning on validation set; may favor simpler models
- ❹ **Single split**
 - Rolling/expanding window validation recommended for robustness
- ❺ **Extreme events**
 - Black swan events remain challenging for all pattern-based models

① Simple models match or beat transformers

- LSTM best at $h = 1$ and $h = 21$
- Ridge best at $h = 5$
- TFT competitive but not superior

② Earnings features add value

- +1.1 pp overall DA improvement
- +7.5 pp during earnings windows (61.4% vs. 53.9%)
- Confirms post-earnings-announcement drift

③ Performance is regime-dependent

- GRU best in bear market (2022)
- LSTM best in bull market (2023–24)
- All models struggle in high-volatility downturns

④ Complexity is not always better

- More parameters $\neq$ better performance
- Data constraints favor simpler models
- Feature engineering remains crucial

What makes this work valuable?

① Rigorous benchmarking

- Apples-to-apples comparison across model classes
- Strict temporal validation (no look-ahead bias)
- Multiple random seeds for statistical reliability

② Honest negative results

- Transformers don't always win
- Important for setting realistic expectations
- Guides future research directions

③ Comprehensive evaluation

- Multi-metric (RMSE, R^2 , DA)
- Regime analysis
- Event window analysis
- Ablations

④ Baseline for future work

1. Multi-Modal Integration

- News, earnings transcripts, social media, macro text
- *Hypothesis:* Transformers excel with heterogeneous data

2. Alternative Data Sources

- Satellite imagery, credit card data, web scraping, order flow

3. Methodological Extensions

- Larger universes (S&P 500, Russell 3000)
- Intraday/high-frequency data; Longer horizons (quarterly, annual)
- Cross-asset learning (stocks, bonds, commodities)

4. Advanced Architectures

- PatchTST, TimesFM-style pretraining, CNN+Transformer hybrids
- Graph neural networks for sector relationships

5. Interpretability & Explainability

- Attention weights, feature importance (SHAP, LIME)
- Regime detection; Understanding when/why models fail

6. Economic Evaluation

- Full backtesting with transaction costs, portfolio optimization
- Risk-adjusted returns (Sharpe, Sortino), real-world deployment

For Practitioners

- 1 Start simple: LSTM/Ridge can match transformers on daily data
- 2 Don't neglect feature engineering
- 3 Earnings events provide predictable opportunities
- 4 Monitor regime changes carefully
- 5 Always validate with strict temporal splits

For Researchers

- 1 Transformers need scale: more data, richer modalities
- 2 Negative results are valuable—publish them!
- 3 Multi-modal integration is promising direction
- 4 Benchmark against strong, well-tuned baselines
- 5 Economic evaluation is crucial next step

Thank You!

Questions?

Contact:

Nelson Siu

`nelson.siu@mail.utoronto.ca`

Jonathan H. Chan

`jonathan@sit.kmutt.ac.th`

Backup: Detailed Performance Table

Model	h	RMSE	R^2	DA	Std
GRU	1	0.016249	-0.006072	0.500920	± 0.024931
GRU	5	0.057883	-0.002885	0.542105	± 0.036308
GRU	21	0.050854	-0.002185	0.539142	± 0.033861
LSTM	1	0.016222	-0.002677	0.516688	± 0.015768
LSTM	5	0.057849	-0.001694	0.554207	± 0.018432
LSTM	21	0.050840	-0.001629	0.550429	± 0.021653
Random Forest	1	0.016387	-0.023148	0.519454	± 0.000290
Random Forest	5	0.039717	-0.206505	0.531379	± 0.000636
Random Forest	21	0.081805	-0.248432	0.542956	± 0.001095
Ridge	1	0.016346	-0.018058	0.500254	± 0.008734
Ridge	5	0.036514	-0.019737	0.511699	± 0.012456
Ridge	21	0.074214	-0.027501	0.528230	± 0.015892
TFT	1	0.016453	-0.031783	0.489379	± 0.019597
TFT	5	0.057907	-0.003690	0.550134	± 0.012030
TFT	21	0.050963	-0.006502	0.538742	± 0.029354
XGBoost	1	0.016464	-0.032850	0.515368	± 0.011245
XGBoost	5	0.038217	-0.117070	0.508611	± 0.014567
XGBoost	21	0.081344	-0.234415	0.516713	± 0.017234

Tabular Models

- **Ridge:** $\alpha \approx 1.0$ (tuned on val)
- **Random Forest:**
 - 100 trees
 - Max depth: 5
 - Min samples leaf: 1
- **XGBoost:**
 - Max depth: 3
 - Learning rate: 0.1
 - 100 rounds

Sequence Models

- **LSTM/GRU:**
 - Hidden units: 64
 - Layers: 1
 - Dropout: 0.2
 - Lookback: 60 days
 - Learning rate: 1×10^{-3}
- **TFT:**
 - Attention heads: 4
 - Layers: 1
 - Hidden dim: 16
 - Dropout: 0.3
 - Learning rate: 3×10^{-4}

Training: Adam/AdamW optimizer, batch size 256, up to 10 epochs with early stopping on validation set

Model	Parameters	Training Time
Ridge	~100	<1 min
Random Forest	~10k	~5 min
XGBoost	~10k	~10 min
LSTM	~25k	~1 hour
GRU	~25k	~1 hour
TFT	~50k	~2–3 hours

Trade-offs:

- Tabular models: fast, competitive accuracy
- Sequence models: moderate cost, best long-horizon performance
- Transformers: expensive, no clear advantage on this dataset

Practical implication: For rapid deployment and iteration, simpler models are advantageous